

Sri Chaitanya IIT Academy.,India.

A.P T.S KARNATAKA TAMILNADU MAHARASTRA DELHI RANCHI

A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

Sec: Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT **Paper -1(Adv-2025-P1-Model)** **Date: 29-03-2026**

Time: 09.00Am to 12.00Pm **GTA-21** **Max. Marks: 180**

29-03-2026_Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT_Jee-Adv(2025-P1)_GTA-21_Syllabus,

Rankers Academy JEE

PHYSICS : TOTAL SYLLABUS

CHEMISTRY : TOTAL SYLLABUS

MATHEMATICS : TOTAL SYLLABUS

Name of the Student: _____ **H.T. NO:** | | | | | | | | |

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JEE-ADVANCED-2025-P1-Model

Time:3Hr's **IMPORTANT INSTRUCTIONS** **Max Marks: 180**

MATHEMATICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 1 – 4)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 5 – 7)	Questions with Multiple Correct Choice with partial mark	+4	-2	3	12
Sec – III(Q.N : 8 – 13)	Questions with NUMERICAL VALUEType	+4	0	6	24
Sec – IV(Q.N : 14 – 16)	Matching Type	+4	-1	3	12
Total				**16**	**60**

PHYSICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 17 – 20)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 21 – 23)	Questions with Multiple Correct Choice with partial mark	+4	-2	3	12
Sec – III(Q.N : 24 – 29)	Questions with NUMERICAL VALUE Type	+4	0	6	24
Sec – IV(Q.N : 30 – 32)	Matching Type	+4	-1	3	12
Total				**16**	**60**

CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 33 – 36)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 37 – 39)	Questions with Multiple Correct Choice with partial mark	+4	-2	3	12
Sec – III(Q.N : 40 – 45)	Questions with NUMERICAL VALUE Type	+4	0	6	24
Sec – IV(Q.N : 46 – 48)	Matching Type	+4	-1	3	12
Total				**16**	**60**

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MATHEMATICS **Max Marks: 60**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : −1 In all other cases

1. Consider the matrix $P=\begin{pmatrix}2&0&0\\0&2&0\\0&0&3\end{pmatrix}$. Let the transpose of matrix X be donoted by X^T. Then the number of 3×3 invertible matrices Q with integer entries, such that is $Q^{-1}=Q^T$ and $PQ=QP,$ is

A) 32 **B)** 8 **C)** 16 **D)** 24

2. The total number of real solutions of the equation $\theta=\tan^{-1}(2\tan\theta)-\frac{1}{2}\sin^{-1}\left(\frac{6\tan\theta}{9+\tan^2\theta}\right)$ is (Here, the inverse trigonometric functions $\sin^{-1}x$ and $\tan^{-1}x$ assume values in $\left[-\frac{\pi}{2},\frac{\pi}{2}\right]$ and $\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$, respectively.)

A) 1 **B)** 2 **C)** 3 **D)** 5

3. Number of solutions of the equation $\cos ec^{-1}(\cos ecx)=\sin x\, in[-2\pi,2\pi]$ is 'a' and number of solutions $\cos ec^{-1}(\cos ecx)=\sqrt{\sin x}\, in[-2\pi,2\pi]$ is 'b' then $\sqrt{a+b}=$

A) an irrational number **B)** a compositive number

C) neither prime nor composite **D)** a prime number

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JEE MAIN 2023 SINGARAJU VENKAT KOUNDINYA RANK 1
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4. Let $\vec{a},\vec{b}$ be two vectors perpendicular to each other and $|\vec{a}|=2,|\vec{b}|=3$ and $\vec{c}\times\vec{a}=\vec{b}$.

When $|\vec{c}-\vec{a}|$ is least then the value of $|\vec{c}|$ is

A) $\frac{1}{2}$ **B)** $\frac{7}{2}$ **C)** $\frac{5}{2}$ **D)** 4

SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **THERE (03)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme :

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Partial Marks: +3 If all the four options are correct but **ONLY** three options are chosen;
Partial Marks: +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct ;
Partial Marks: +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option ;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : –2 In all other cases.

For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then choosing ONLY (A), (B) and (D) will get +4 marks; choosing ONLY (A) and (B) will get +2 marks; choosing ONLY (A) and (D) will get +2 marks; choosing ONLY (B) and (D) will get +2 marks; choosing ONLY (A) will get +1 mark; choosing ONLY (B) will get +1 mark; choosing ONLY (D) will get +1 mark; choosing no option (i.e. the question is unanswered) will get 0 marks; and choosing any other combination of options will get -2 marks.

5. The median and mode of the following wage distribution of 230 gig workers are known to be Rs. 33, 500 and Rs. 34,000 respectively

Wages (in '000 Rs)	0-10	10-20	20-30	30-40	40-50	50-60	60-70
Frequency	4	16	f_3	f_4	f_5	6	4

Which of the following statements is(are) correct?

A) $f_3=60$ **B)** $f_4=100$ **C)** $f_5=40$ **D)** $f_5=50$

6. Let $S(n,k)$ denote the number of ways in which a distinct balls can be put into k identical boxes such that no box remains empty $(k\le n)$. Which of the following is(are) correct?

A) $S(5,3)=25$

B) $S(6,3)=S(5,3)+S(5,2)$

C) $S(6,3)=3S(5,3)+S(5,2)$

D) The number of onto function from 6 element set to 3 element set is $3!S(6,3)$

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7. Let P be a point on the ellipse $\frac{x^2}{25}+\frac{y^2}{9}=1$, in first quadrant with focal distance 7 units w.r.t focus $S'(-4,0)$. Let $S'P$ and perpendicular from focus S on the tangent at P intersect at Q. Then which of the following is not TRUE?

A) PQ is equal to 3

B) SQ is equal to $\frac{3\sqrt{3}}{\sqrt{7}}$

C) Slope of tangent at P is equal to $\frac{\sqrt{3}}{5}$

D) Let P be a variable point then locus of Q is a circle with radius=10 units.

SECTION-III

(NUMERICAL VALUE)

This section contains **SIX (06)** questions.

- The answer to each question is a NUMERICAL VALUE.
- For each question, enter the correct integer corresponding to the answer using the mouse and the onscreenvirtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme

Full Marks :+4 If ONLY the correct integer is entered;Zero Marks : 0 In all other cases.

8. Consider two distinct polynomials $f(x)$ and $g(x)$ of degree less than or equal to 3 with integral coefficients such that

$f(1)=g(1), f(-1)=-g(-1), f(2)=g(2), f(-2)=-g(2), f(3)=g(3), f(-3)=-g(-3)$.

Find the minimum value of $f^2(0)+g^2(0)$?

9. Find the ratio of number of obtuse angled triangles to the number of acute angled triangle formed by joining the vertices of a 12 sided regular convex polygon.

10. Let $\left(x^3+px^2+2x-5\right)^{19}\left(x^2+qx-41\right)^8\left(x^4-x^3+x-7\right)^6$

$=x^{97}+391x^{96}+a_{95}x^{95}+a_{94}x^{94}+.....a_1x+a_0$ be an identity, where $p,q,a_{95}....a_0$ are integers. Compute the smallest positive value of p.

11. A person tosses a fair coin n times (n is not a multiple of 3 and $n\geq 4$). He wins if he is able to get heads in multiple of 3 times. The probability that he wins given that he gets head atleast one time is given by $P(n)$. Then $126P(6)$ equals _____.

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12. If m is the minimum value of $f(x)$, where $f(x)=x^{3/2}+\frac{1}{8x^{3/2}}-\frac{17}{4}\left(x+\frac{1}{4x}\right), x>0$,

then the value of $100+2m$ is _____

13. Let $a_0=-2,\ b_0=1$ and for $n\in w$, ($w=$ set of whole numbers)

$$a_{n+1}=a_n+b_n+\sqrt{a_n^2+b_n^2}$$

$$b_{n+1}=a_n+b_n-\sqrt{a_n^2+b_n^2}$$

If $a_{2012}=2^m\sqrt{2^n+2}-2^{2011}$, Find $2m-n=$_____

SECTION – IV
(MATCHING TYPE)

This section contains **THERE (03)** Matching List Sets.

- Each set has **ONE** Multiple Choice Question.
- Each set has **TWO** lists :**List-I** and **List-II.**
- **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Five** entries (P), (Q), (R), (S) and (T).
- **FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks:+3 **ONLY** if the option corresponding to the correct combination is chosen;
Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks: –1 In all other cases.

14.

	COLUMN-I		COLUMN-II
P)	Let a,b,c are positive real numbers such that $a^3b^2c=12$, then the minimum value of $49a+3b+c$ is equal to	**1)**	1
Q)	The minimum value of $\left\|2x^3-\frac{3}{x^2}\right\|$ for $x<0$ is equal to	**2)**	5
R)	The maximum value of $\frac{x^5\left(8-x^3\right)}{\sqrt[3]{25}}$ for $0<x<2$ is equal to	**3)**	7
S)	If $x^7y^5=a$ and $7x+5y\geq 12\forall x,y>0$, then the minimum value of 'a' is equal to	**4)**	15
		5)	42

A) P - 3; Q - 3; R- 4; S -1 **B)** P - 5; Q - 3; R- 2; S -4
C) P - 4; Q - 2; R- 3; S - 1 **D)** P - 5; Q - 2; R- 4; S - 1

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15. If $f(x)=\begin{vmatrix}(\alpha x+1)\cos^2 x & x & 1-x\\ \beta\sin x & x^2 & 2x\\ (\gamma x^2+1)\tan x & x & 1-x^2\end{vmatrix}$

	COLUMN-I		COLUMN-II
P)	$\lim\limits_{x\to 0}\dfrac{f(x)}{x}+\lim\limits_{x\to 0}\dfrac{f(x)}{x^2}$	**1)**	0
Q)	$\lim\limits_{x\to 0}\dfrac{f'(x)}{x}+f'(0)$	**2)**	-1
R)	If $\lim\limits_{x\to 0}\dfrac{1}{x^6}\int\limits_{x^2}^{x^3} f(t)dt=A$, then the value of $[A]$ is ([.] denotes greatest integer function)	**3)**	-2
S)	If $\alpha=\beta=\gamma=0, g(x)=\dfrac{f(x)}{x^2}$ then $\left[g\left(\dfrac{\pi}{4}\right)\right]$ is ([.] denotes greatest integer function)	**4)**	2
		5)	1

A) P - 1; Q - 3; R- 2; S - 1 **B)** P - 2; Q - 3; R- 3; S - 1
C) P - 2; Q - 3; R- 1; S - 1 **D)** P - 1; Q - 5; R- 4; S - 2

16. If an equilateral triangle ABC with vertices at z_1, z_2 and z_3 be inscribed in the circle $|z|=2$ and again a circle is inscribed in the triangle ABC touching the sides AB, BC and CA at $D(z_4), E(z_5)$ and $F(z_6)$ respectively:

	COLUMN-I		COLUMN-II
P)	Then value of $\text{Re}(z_1\bar{z}_2+z_2\bar{z}_3+z_3\bar{z}_1)$ is equal to	**1)**	2
Q)	If $\dfrac{4z_1}{z_3}$ is equal to $a(-1+i\sqrt{3})$, then a is	**2)**	-6
R)	The value of $\lvert z_1+z_2\rvert^2+\lvert z_2+z_3\rvert^2+\lvert z_3+z_1\rvert^2$ is	**3)**	12
S)	If P is any point on incircle the value of $DP^2+EP^2+FP^2$ is	**4)**	6
		5)	-2

A) P - 3; Q - 2; R- 5; S - 1 **B)** P - 1; Q - 2; R- 3; S - 4
C) P - 2; Q - 1; R- 5; S - 4 **D)** P - 2; Q - 1; R- 3; S - 4

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PHYSICS **Max Marks: 60**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
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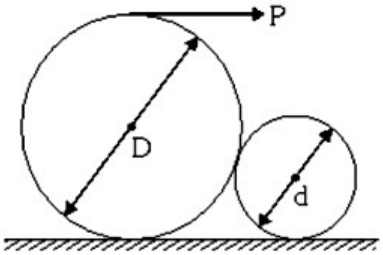
17. A conducting rod of mass m and resistance R is placed on a pair of parallel, smooth conducting rails of negligible resistances separated by a distance L. The rails are inclined at an angle θ to the horizontal. A uniform magnetic field of strength B is directed vertically upwards. The rod is released from rest and slides down the rails. Find terminal velocity (v_t) of the rod.

A) $\frac{mgR\tan\theta}{B^2L^2}$ **B)** $\frac{mgR\sin\theta}{B^2L^2}$ **C)** $\frac{mgR\tan\theta}{B^2L^2\sin\theta}$ **D)** $\frac{mgR\tan\theta}{B^2L^2\cos\theta}$

18. In a setup of photoelectric experiment, a light of intensity $384 W/m^2$ is used which is equally distributed among three wavelengths 620 nm, 496 nm and 310 nm. The light is incident perpendicularly on a surface of area 0.01cm^2, having work function of 2.25eV. Assuming there is no loss of light and efficiency of emission of electrons from the surface is 50%. The photocurrent in the circuit is

A) 83.2×10^{-6} amp **B)** 166.4×10^{-6} amp
C) 20.8×10^{-6} amp **D)** 41.6×10^{-6} amp

19. A heavy cylindrical roller of diameter D and a light cylindrical roller of diameter d rest on horizontal plane as shown in figure.



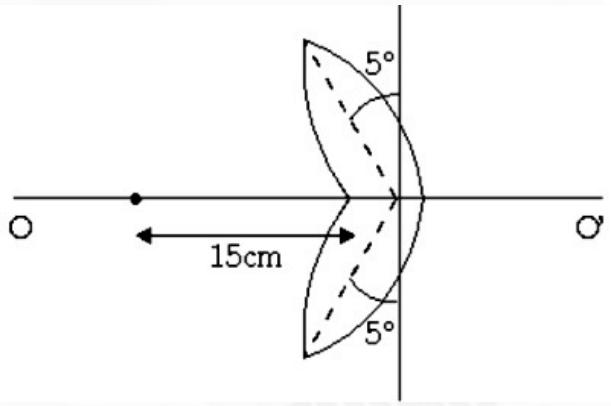
The heavy roller has a string wound round it to which a horizontal force P can be applied as shown in figure. Assuming that the coefficient of friction μ has the same value for all surfaces of contact, determine the limits of μ so that the larger roller can be pulled over the smaller one. (Assume that smaller sphere remains stationary during this process).

A) $\mu\geq\sqrt{\frac{d}{D}}$ **B)** $\mu<\sqrt{\frac{d}{D}}$ **C)** $\mu\geq\sqrt{\frac{2d}{D}}$ **D)** $\mu\geq\sqrt{\frac{d}{2D}}$

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20. A convex lens of focal length 10 cm is cut along its diameter and two halves are joined together such that the angle between their planes is 170°. They are placed symmetrically with respect to the axis OO' as shown in figure.



To the left of lens, a point object is placed at a distance of 15cm from the lens system. The position and separation between the images formed by the lens system:

A) Nearly 30 cm right from the lens and both images do not coincide.

B) Nearly 30 cm right from the lens and both images coincide.

C) Nearly 45 cm right from the lens and both images do not coincide.

D) Nearly 45 cm right from the lens and both images coincide.

SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **THERE (03)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
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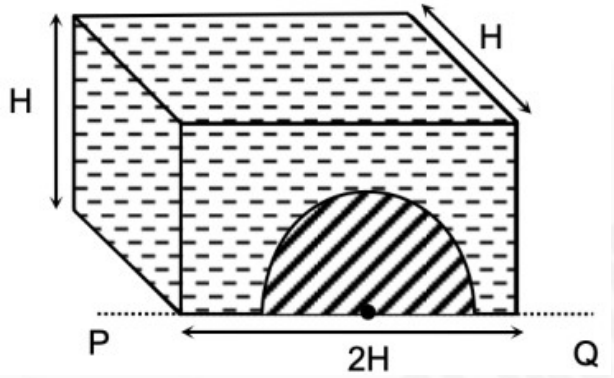
Full Marks :+4 **ONLY** if (all) the correct option(s) is(are) chosen;
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Partial Marks: +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct ;
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Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
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For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then choosing ONLY (A), (B) and (D) will get +4 marks; choosing ONLY (A) and (B) will get +2 marks; choosing ONLY (A) and (D) will get +2 marks; choosing ONLY (B) and (D) will get +2 marks; choosing ONLY (A) will get +1 mark; choosing ONLY (B) will get +1 mark; choosing ONLY (D) will get +1 mark; choosing no option (i.e. the question is unanswered) will get 0 marks; and choosing any other combination of options will get -2 marks.

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21. A liquid of density 'ρ' is filled in a cuboidal container of dimension $(H \times H \times 2H)$ as shown in the figure.



A semicircle gate is made at the bottom of container of radius $R = \frac{H}{2}$. Let F is the force acting on the gate due to the liquid and τ is the torque acting on the gate due to the liquid about a line PQ. Then choose the correct option(s).

A) $F = \frac{H^3 \rho g}{4}\left(\frac{\pi}{4} - \frac{1}{3}\right)$ **B)** $F = \frac{H^3 \rho g}{2}\left(\frac{\pi}{4} - \frac{1}{6}\right)$

C) $\tau = \frac{H^4 \rho g}{4}\left(\frac{1}{3} - \frac{\pi}{32}\right)$ **D)** $\tau = \frac{H^4 \rho g}{3}\left(\frac{1}{3} - \frac{\pi}{16}\right)$

22. Consider a hypothetical hydrogen like atom. The wavelength of spectral line from p^{th} orbit to ground state is given by $\lambda = \frac{3500 p^2}{p^2 - 1}$ in $\mathring{A}$. Choose the correct option(s) for considered hypothetical atom. $(hC = 1240\,\text{eV} - \text{nm})$

A) The ionization potential of considered hypothetical atom is 3.54 V approximately.

B) The ionization potential of considered hypothetical atom is 0.354 V approximately.

C) The wavelength of least energetic photon in the spectral line series is 466.67 nm approximately.

D) The wavelength of maximum energetic photon in the spectral line series is 350.00 nm approximately.

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23. In a Searle's experiment, a wire of diameter $d = 0.060$ cm is used to find its young's modulus. The length of given wire is $L = 200$ cm and when weight of mass $m = 30.0$ kg is used in experiment, extension in the length of wire was found to be 0.200 cm. Then choose the correct option(s).

A) Maximum permissible error in young's modulus of wire is 4.67%.

B) Maximum permissible error in young's modulus of wire is 3%.

C) The least count of the instrument by which length of wire is to be measured is 1 cm.

D) The least count of the instrument by which diameter of the wire is to be measured is 0.001 cm.

SECTION-III

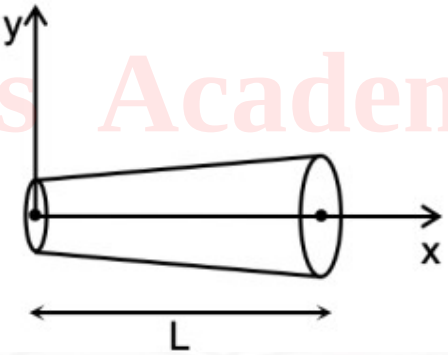
(NUMERICAL VALUE)

This section contains **SIX (06)** questions.

- The answer to each question is a NUMERICAL VALUE.
- For each question, enter the correct integer corresponding to the answer using the mouse and the onscreenvirtual numeric keypad in the place designated to enter the answer.
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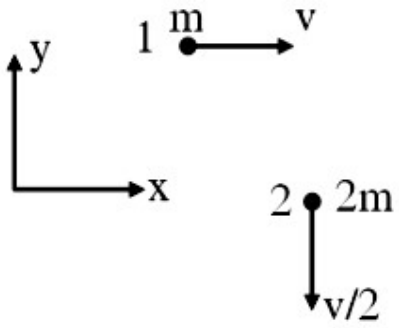
24. A rope of length 'L' is kept along $x-$axis such that its one end is at the origin as shown in the figure.



An external force is applied on rope such that restoring force developed in the rope is given by $F_0\left(1+\frac{x}{L}\right)$, where F_0 is constant. The young's modulus of the rope is given by $Y_0\left(1+\frac{x}{L}\right)^{-1}$ and cross-sectional area is $A_0\left(1+\frac{x}{L}\right)$, where Y_0 and A_0 are constant. The elongation in the rope is $\Delta\ell$ mm. It is given that $F_0 = 1\text{kN}$, $Y_0 = 2\times10^{11}\,\text{N/m}^2$, $A_0 = 5\,\text{mm}^2$ and $L = 1\text{m}$. The value of $\Delta\ell$ is _____.

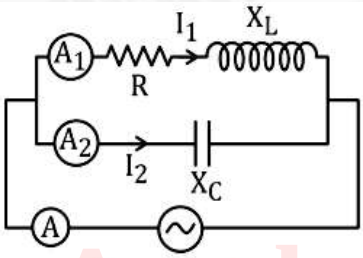
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25. Two particles, one of mass m and velocity v, and another of mass 2m and velocity $v/2$, move perpendicularly on a smooth horizontal surface as shown in figure.



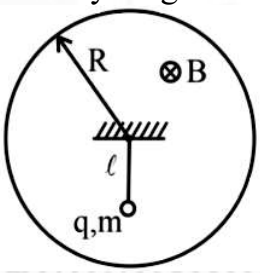
At a certain instant, forces of equal magnitude (and same direction) act on these particles for same time. When these forces stop acting, the first particle acquires a motion perpendicular to its initial direction, with the same speed in $+y$ direction. The magnitude of the velocity acquired by the second particle is $\frac{\alpha v}{2}$, the value of α is

26. Net current will be at angle θ with I_1, V_0 is rms voltage of source. Find θ (in degree).



$X_L = R = 4\ \Omega\ ; X_C = 8\ \Omega\ ; V_0 = 20\sqrt{2}$ v

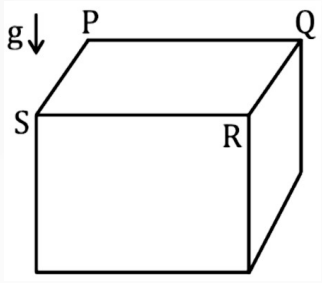
27. A pendulum of mass m and charge q is hanging by a string of length $\ell = 2\text{m}(\ell < R)$ from centre of magnetic field created by long solenoid as shown in figure,



having n turns/length $(n = 500)$ and current $i = \alpha t$ Neglect magnetic field due to motion of charge of bob $\left(\text{Given}\,\alpha = \pi A/\text{sec}, m = 0.4\,\text{gm}, g = 10\,\text{m/s}^2,\right.$ $\left.\pi^2 = 10, \mu_0 = 4\pi \times 10^{-7}\,\text{S.I.unit}, q = 1C, 3^{1/4} = 1.32\right)$. Find equilibrium angle in degree of pendulum with vertical.

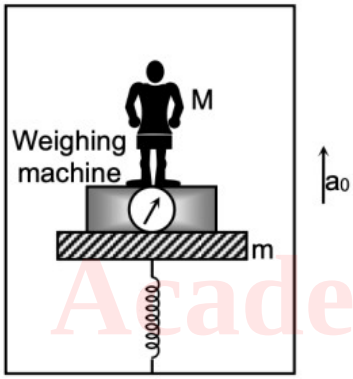
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28. A cube of mass m and side length a is placed as shown in a region of magnetic field $\vec{B} = B_0\hat{i} - B_0t^2\hat{j}$. A square conducting loop $PQRS$ is attached to the top face of the cube. 'r' is the resistance of the loop.



Then the time at which cube topples is $\frac{mgr}{kB_0^2a^3}$. Then $k =$ _____ $\left(\vec{g} = -g\hat{j}\right)$

29. An elevator is moving vertically upward with an acceleration $a_0 = 4\,\text{m/s}^2$. A man of mass $M = 60$ kg is standing on a massless weighing machine which is kept on a platform of mass $m = 10$ kg. Platform is connected with a spring as shown in the figure.



Platform oscillates with a frequency of $0.5\,\text{sec}^{-1}$ and amplitude 0.2 m. N_0 is the maximum reading of the weighing machine in kilogram. Find N_0. $\left(\text{Take}\,\pi^2 = 10, g = 10\text{m/s}^2\right)$

SECTION – IV
(MATCHING TYPE)

This section contains **THERE (03)** Matching List Sets.
- Each set has **ONE** Multiple Choice Question.
- Each set has **TWO** lists :**List-I** and **List-II.**
- **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Five** entries (P), (Q), (R), (S) and (T).
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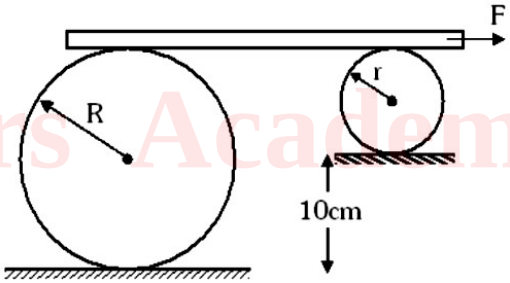
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30. An air column in a pipe closed at one end is vibrating in 2nd overtone due to tuning fork of frequemcy 660 Hz. The speed of sound in air 330 m/s. $L(m)$ is the length of tube.

	COLUMN-I		COLUMN-II
I)	Ratio of pressure amplitude at $\frac{L}{10}$ from open end, and at the centre of tube.	**P)**	1
II)	If difference of phase of oscillating particle at distance $\frac{L}{10}$ and $\frac{L}{2}$ from open end is $\frac{n\pi}{6}$, then n is	**Q)**	0
III)	Length of the tube is L then $8L$ is	**R)**	5
IV)	If atmospheric pressure is P_0, sum of the maximum and minimum pressure at closed end is mP_0, then m is	**S)**	2
		T)	6

A) I-T; II-P; III-R; IV-S **B)** I-P; II-T; III-S; IV-R
C) I-P; II-T; III-R; IV-S **D)** I-P; II-Q; III-R; IV-S

31. A heavy plank of mass 102.5 kg is placed over two solids cylindrical rollers of radii $R = 10$ cm and $r = 5$ cm. Mass of roller are 40 kg and 20 kg respectively. Plank is pulled towards right by applying a horizontal force $F = 25$ N as shown in figure.



During first second of motion the plank gets displaced by 10 cm. If plank remain horizontal and slipping does not take place, calculate magnitude of force of friction acting between in N.

	COLUMN-I		COLUMN-II
I)	Plank and bigger roller	**P)**	0.5
II)	Plank and smaller roller	**Q)**	3
III)	Bigger sphere and floor	**R)**	1.5
IV)	Smaller sphere and floor	**S)**	1
		T)	10

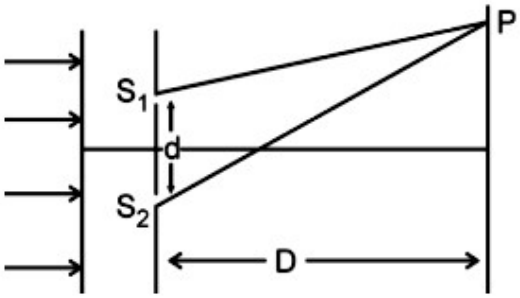
A) I-P; II-Q; III-T; IV-R **B)** I-Q; II-S; III-R; IV-P
C) I-Q; II-R; III-S; IV-P **D)** I-P; II-R; III-S; IV-P

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32. Plane wave-front in a YDSE experiment, wavelength of light

$\lambda = 6000\,\text{Å}, D = 120\,\text{cm}, d = 1.2\,\text{mm}$, both sources S_1 and S_2 produces intensity I_0.



	COLUMN-I		COLUMN-II
I)	A slab of thickness $t_1 = 0.02\text{mm}$, $\mu_1 = 1.6$ is placed before S_1 and a slab of thickness $t_2 = 0.03\text{mm}$ and $\mu_2 = 1.4$ before S_2, find net shift in central maxima on screen (in nm)	**P)**	400
II)	Minim thickness of slab $\mu = 1.5$ placed on S_2 to make intensity at centre of screen equal to zero (in nm)	**Q)**	500
III)	If intensity of point P is 0.75 times of intensity of central maxima then minimum path difference corresponding to point $'P'$ is (in nm)	**R)**	100
IV)	If screen translates due right with 10 cm/s then after 10 sec the change in width of central maxima is $1000x$ nm, then x is:	**S)**	600
		T)	0

A) I-T; II-S; III-R; IV-Q **B)** I-T; II-S; III-Q; IV-R
C) I-S; II-T; III-R; IV-Q **D)** I-S; II-T; III-Q; IV-R

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CHEMISTRY **Max Marks: 60**

SECTION – II
(SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

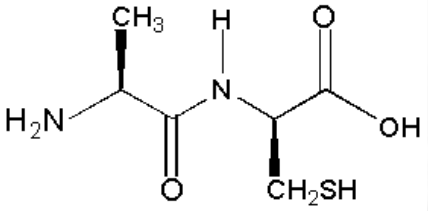
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

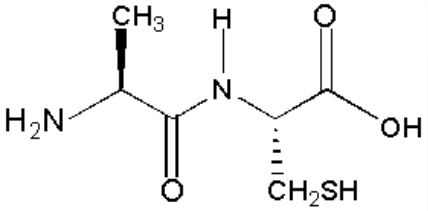
Full Marks : +3 If **ONLY** the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : −1 In all other cases

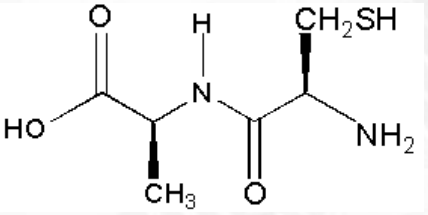
33. Upon the heating Cu_2S the reactant that does not give copper metal is

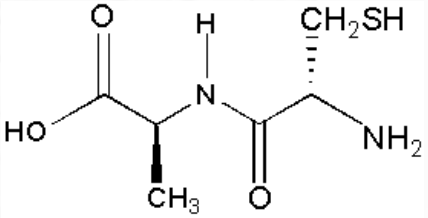
A) $CuFeS_2$ **B)** CuO **C)** Cu_2O **D)** $CuSO_4$

34. The correct formula of the natural dipeptide $Ala-Cys$ is

A) 

B) 

C) 

D) 

35. In the reaction $N_2(g)+3H_2(g) \rightleftharpoons 2NH_3(g)+Heat$. One mole of N_2 reacts with three moles of H_2. If at equilibrium 'x' moles of N_2 have reacted, then the value of 'x' in terms of K_P of the above reaction and the total pressure of gases at equilibrium (P) is $(\text{assume } x << 1)$

A) $\frac{3P.\sqrt{3.K_P}}{8}$ **B)** $\frac{8P}{3\sqrt{3.K_P}}$ **C)** $\frac{3\sqrt{3.K_P}}{8P}$ **D)** $\frac{8}{P}.\sqrt{\frac{K_P}{3}}$

36. Correct statement of the following is

A) Maximum permissible concentration of nitrate in drinking water is 70 ppm

B) In world war – II pesticide used to control malaria is "Lewisite"

C) Hydroxy apatite is much harder than fluorapatite

D) Clean water has BOD value less than 5 ppm

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SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

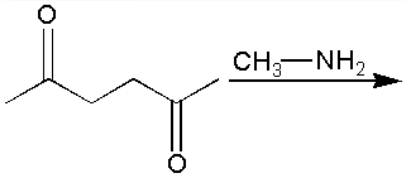
This section contains **THERE (03)** questions.

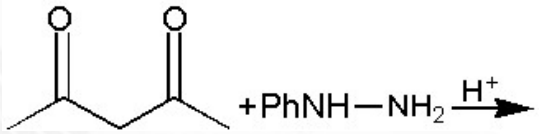
- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme:

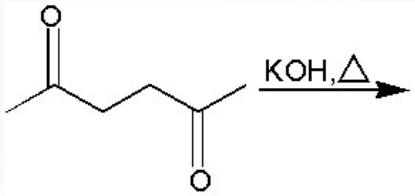
Full Marks: +4 **ONLY** if (all) the correct option(s) is(are) chosen;
Partial Marks: +3 If all the four options are correct but **ONLY** three options are chosen;
Partial Marks: +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct;
Partial Marks: +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option;
Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks: –2 In all other cases.

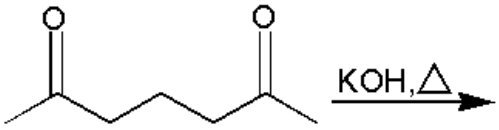
For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then choosing ONLY (A), (B) and (D) will get +4 marks; choosing ONLY (A) and (B) will get +2 marks; choosing ONLY (A) and (D) will get +2 marks; choosing ONLY (B) and (D) will get +2 marks; choosing ONLY (A) will get +1 mark; choosing ONLY (B) will get +1 mark; choosing ONLY (D) will get +1 mark; choosing no option (i.e. the question is unanswered) will get 0 marks; and choosing any other combination of options will get -2 marks.

37. In which of the following reaction (s) product is/are aromatic

A) 

B) 

C) 

D) 

38. Which of the following statement(s) is/are **incorrect**?

A) Combustion of methane gas in an adiabatic rigid container will cause no change in internal energy of the system but will increase temperature of the system.

B) It is possible to have both adiabatic reversible and adiabatic irreversible processes involving ideal gases between the same initial and final states.

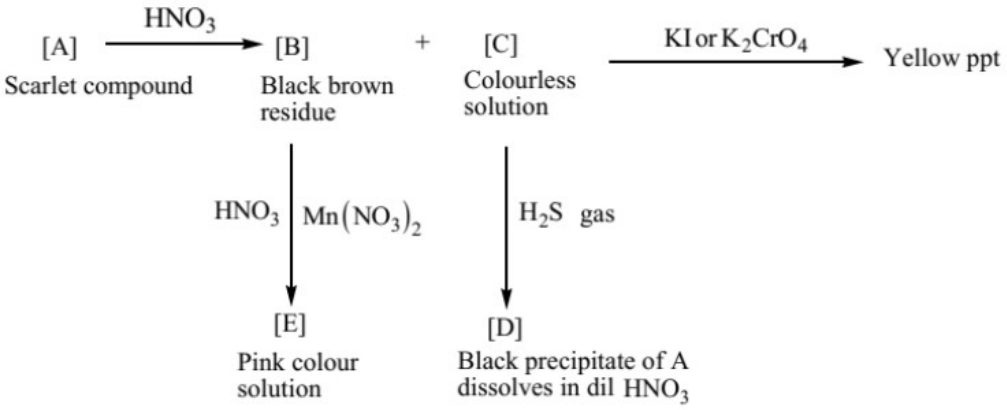
C) For a chemical reaction involving only ideal gases and occurring at constant temperature, there will not be any change in the internal energy of the system and reactions will be driven by an increase in entropy of the system.

D) P – V work is always non-zero when there is an increase in the volume of system during the process.

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39. Consider the following reaction schemes.



A) The black precipitate (D) is PbS

B) The pink color solution (E) is due to the formation of $HMnO_4$

C) Reaction of A to form B and C is redox reaction

D) Reaction of B to form E is redox reaction

SECTION-III
(NUMERICAL VALUE)

This section contains **SIX (06)** questions.

- The answer to each question is a NUMERICAL VALUE.
- For each question, enter the correct integer corresponding to the answer using the mouse and the onscreenvirtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme

Full Marks :+4 If ONLY the correct integer is entered;Zero Marks : 0 In all other cases.

40. A complex is formed with the formula $\left[M(AB)a_2bc\right]$ which shows

Possible no. of enantiomer pair = 'p'

Possible no. of geometrical isomers = 'q'

Then find $(p \times q)$?

41. Among the following complexes, how many have spin only magnetic moment of 2.83 BM?

$$\left[Ni(CO)_4\right];\left[Ni(CN)_4\right]^{2-};\left[NiF_6\right]^{2-};\left[NiF_6\right]^{4-};\left[Ni(en)_3\right]^{2+};\left[Ti(H_2O)_6\right]^{2+};\left[V(NH_3)_6\right]^{3+}$$

42. $$\text{Cycloheptene} \xrightarrow{KMnO_4/H^+} X \xrightarrow[II)\Delta]{I)Ca(OH)_2} Y \xrightarrow[\text{II) Conc.}H_2SO_4]{I)NH_2OH,\ pH=4-5} Z$$

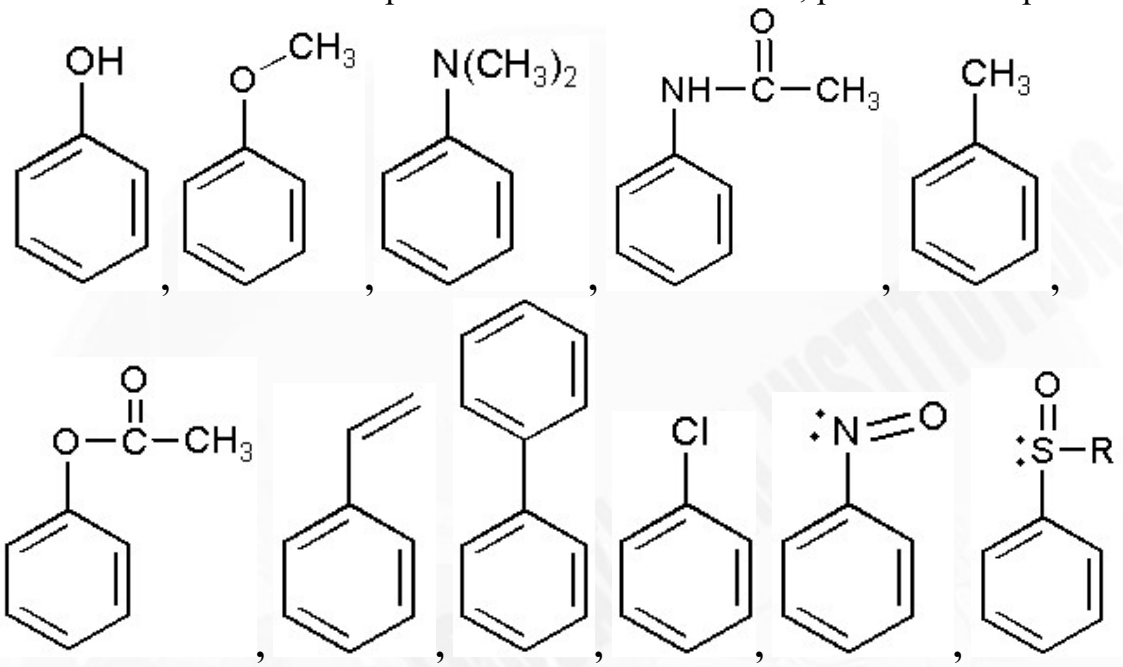
$$Z \xrightarrow[H_2O]{533-543K} (P)_n$$

Gram formula mass (in g) of P is (At wt of C is 12, H is 1, N is 14, O is 16).

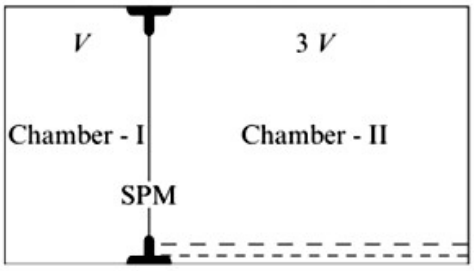
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43. Among the following how many substituted benzene derivatives are more reactive than benzene in aromatic electrophilic substitution at all ortho, para and meta positions?



44. A container is divided into two parts in the ratio 1 : 3, by a fixed selectively permeable membrane (SPM), which permits the movement of CO gas only. At $T\,K$, Chamber – I contains only $COCl_2$ gas at 20 mm Hg pressure and Chamber – II contains small amount of a liquid 'X', but is sufficient to maintain equilibrium with vapours at any temperature. Now, both the chambers are heated to '$2T$'K at which 50% of $COCl_2$ is dissociated into CO and Cl_2. The vapour pressure of $X(l)$ at $T\,K$ and $2T\,K$ are 20 and 30 mm Hg, respectively. What is the magnitude of pressure difference (in mm of Hg) between the two chambers at $2T\,K$.



45. Percentage of ammonium chloride hydrolysed in its 0.1 M solution at 25°C is $x\times10^{-2}$. Find x.

$E_{\substack{NH_4Cl|H_2\\0.025M\ \ 1bar}} = -0.3V, 2.303RT/F = 0.06$

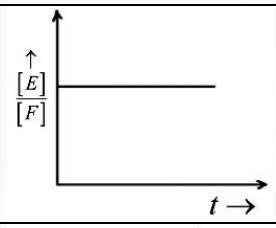
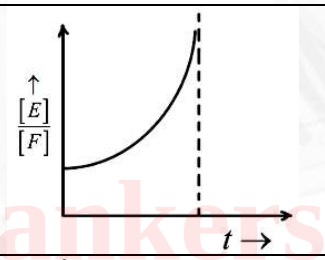
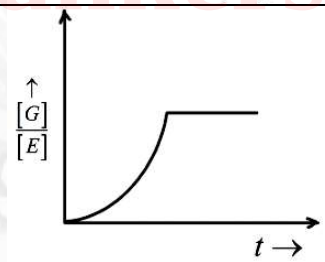
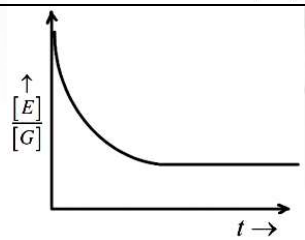
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SECTION – IV
(MATCHING TYPE)

This section contains **THERE (03)** Matching List Sets
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Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks: –1 In all other cases..

46. For the zero order gas phase reaction, $E(g)+F(g)\rightarrow G(g)$, graphs A to D are plotted for different experiments.

COLUMN-I **Mole ratios vs Time for different experiments**		**COLUMN-II** **Conclusion from graph**	
I)		**P)**	Intercept on y-axis=1
II)		**Q)**	Intercept on y-axis<1
III)		**R)**	E is the limiting reagent
IV)		**S)**	F is the limiting reagent

A) I - P ; II - S; III - QS; IV - S
B) I - P; II - Q; III - R; IV - S
C) I - R; II - S; III - R; IV - P
D) I - P; II - S; III - PR; IV – S

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47. List – I contains elements and List – II contains increasing order of indicated properties.

List – I		List – II	
(I)	N, F, O, Li	(P)	Increasing order of $\lvert EA \rvert$
(II)	F, N, Si, Mg, Rb	(Q)	Increasing order of IE_1
(III)	K, Na, Al, Mg	(R)	Increasing order of IE_2
(IV)	N, B, O, Cl	(S)	Increasing order of radius

A) I – R, II – Q, III – S, IV – P

B) I – S, II – R, III – P, IV – Q

C) I – R, II – S, III – Q, IV – P

D) I – S, II – R, III – Q, IV – P

48. Match **List I** with **List II**

Column I		Column II	
P)	C_6H_5CHO	**I)**	Can reduce Tollen's reagent
Q)	D – Fructose	**II)**	Can give red precipitate with fehlings solution
R)	2 – Butene	**III)**	Can reduce bromine water
S)	1 – Butyne	**IV)**	Can decolourise Bayer's reagent

A) P-I,II; Q-I,II,III; R-III,IV; S-I,II,IV

B) P-I,II; Q-I,II; R-IV, S-I,IV

C) P-I; Q-I,II,III; R-III,IV; S-II,III,IV

D) P-I; Q-I,II; R-III,IV; S-III, IV

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16 RANK DEVDUTTA MAJHI HT. No. 255053116*	18 RANK DHARMANA GNANA RUTVIK SAI HT. No. 256055278	19 RANK VANGALA AJAY REDDY HT. No. 256131009	23 RANK AKSH GOGI HT. No. 252071075*	26 RANK P HEMA SAI SURYA KARTHIK HT. No. 256033006	27 RANK SARKARSAMUDRA HT. No. 252071105*
30 RANK OM PRAKASH BEHERA HT. No. 252021018*	32 RANK SUNKARA SAI RISHANTH REDDY HT. No. 256165327	34 RANK DHRUBA JYOTHI PANJA HT. No. 252048248*	35 RANK BHAVESH JAYANTHI HT. No. 251043080	36 RANK ADVAY MAYANK HT. No. 252104113*	37 RANK KARMANYA GUPTA HT. No. 252081477*
42 RANK MD ANAS HT. No. 252046210*	45 RANK RAMIT GOYAL HT. No. 257001113*	52 RANK MAULIK JAIN HT. No. 252079407*	54 RANK GARV HT. No. 252056188*	59 RANK LARISSA HT. No. 252079071*	60 RANK ARYAN BALABADRULA HT. No. 256132077
63 RANK SAMYAJYOTI BISWAS HT. No. 255058456*	64 RANK AARUSH ANAND HT. No. 251006176*	72 RANK RUSHMITH BANDARI HT. No. 256168048	78 RANK KORIKANA RASAGNYA HT. No. 256057046	87 RANK LAKSHYA SHARMA HT. No. 252070079*	91 RANK AVANEESH BANSAL HT. No. 251113130*	95 RANK KAVYA AGGARWAL HT. No. 252079121*

BELOW 100 ALL INDIA OPEN CATEGORY RANKS: 29	BELOW 500 ALL INDIA OPEN CATEGORY RANKS: 113	BELOW 1000 ALL INDIA OPEN CATEGORY RANKS: 205	BELOW 1000 ALL INDIA CATEGORY RANKS COUNT: 745	NUMBER OF QUALIFIED RANKS: 4,212

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